# 1. Set notation and basic counting

**Combinatorics**

**MATH 31003-01 - Fall 2026**

Combinatorics is all about counting objects. We introduce the notation for ”containers” of objects a.k.a. **sets** and define operations on sets. Next, we study the first two counting methods: the sum rule and product rule.

## 1.1 Set notation

A **set** is a collection of distinct objects. We usually denote a set by a capital letter, such as $A$, and an element of a set by a lowercase letter, such as $a$. The notation

$$a \in A$$

means that $a$ is an element of $A$, while $a \notin A$ means that it is not. We can describe a set by listing its elements surrounded by braces.

**Example 1.1.1** (The decimal-digit set). $D = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ is the set of decimal digits.

Because a set records membership rather than order, $\{1, 2, 3\} = \{3, 2, 1\}$. Repeating an element also does not change a set: $\{1, 1, 2\} = \{1, 2\}$. We can also describe a set by a property its elements satisfy. If $U$ is a specified **universe** of possible elements, then we use the notation

$$A := \{a \in U : a \text{ satisfies the stated property}\}.$$

The colon is read as ”such that”.

**Example 1.1.2** (Sets described by a property).

- $E := \{n \in \mathbb{Z} : n \text{ is even}\}$ is the set of even integers, where $\mathbb{Z}$ denotes the integers.
- If $U := \{1, 2, 3, 4, 5, 6\}$, then $A := \{x \in U : x > 3\} = \{4, 5, 6\}$.

The symbol $\emptyset$ denotes the **empty set**, the set with no elements.

### 1.1.1 Set operations

Let $A$ and $B$ be subsets of the same universe.

- The **intersection** of $A$ and $B$ consists of the elements belonging to both sets:

$$A \cap B := \{x : x \in A \text{ and } x \in B\}.$$

- The **union** of $A$ and $B$ consists of the elements belonging to at least one of the sets:

$$A \cup B := \{x : x \in A \text{ or } x \in B\}.$$

Here ”or” is inclusive: an element belonging to both $A$ and $B$ belongs to their union.

- The **set difference**, also called the **relative complement**, consists of the elements of $X$ that do not belong to $A$:

$$X \setminus A := \{x \in X : x \notin A\}.$$

- The **Cartesian product** of two sets $A$ and $B$ is the set of *ordered pairs* $(a, b)$ with $a$ chosen from $A$ and $b$ chosen from $B$:

$$A \times B := \big\{(a, b) : a \in A \text{ and } b \in B\big\}.$$

For more than two sets we can write $A \times B \times C \dots$. If all the sets are the same we write $A^2 := A \times A$ and so on.

- Two sets are **disjoint** if they have no elements in common; equivalently, $A \cap B = \emptyset$.

**Example 1.1.3** (Operations on two finite sets). If $A = \{1, 2, 3\}$ and $B = \{3, 4\}$, then

$$\begin{aligned} A \cap B &= \{3\}, & A \cup B &= \{1, 2, 3, 4\}, \\ A \setminus B &= \{1, 2\}, & A \times B &= \Big\{(1,3), (1,4), (2,3), (2,4), (3,3), (3,4)\Big\}. \end{aligned}$$

### 1.1.2 Cardinality and inclusion

For a finite set $A$, its **cardinality** $|A|$ is the number of elements in $A$.

**Example 1.1.4** (Cardinalities of finite sets). $|\{2, 4, 6, 8\}| = 4$ and $|\emptyset| = 0$.

We write $A \subseteq B$ if every element of $A$ is also an element of $B$:

$$A \subseteq B \quad \iff \quad \text{for every } x,\ x \in A \text{ implies } x \in B.$$

We write $A \subsetneq B$ if $A \subseteq B$ but $A \neq B$. Two sets are equal precisely when they have the same elements:

$$A = B \quad \iff \quad \text{for every } x \in A \text{ if and only if } x \in B.$$

Thus, to prove $A = B$, it is enough to prove both $A \subseteq B$ and $B \subseteq A$.

### 1.1.3 Logical quantifiers and summation notation

The symbol $\forall$ means "for every". The symbol $\exists$ means "there exists".

**Example 1.1.5** (Reading quantifiers).

- $\forall x \in A, \quad x \geq 0$ means that every element of $A$ is nonnegative.
- $\exists x \in A$ such that $x^2 = 4$ says that at least one element of $A$ has square 4.

The symbol $\exists!$ means "there exists a unique." Thus $\exists! x \in A$ such that $P(x)$ asserts two things: at least one element of $A$ satisfies $P$, and no two different elements of $A$ both satisfy $P$.
The notation

$$\sum_{j=1}^{n} a_j$$

means that we add the terms $a_j$ as the integer index $j$ runs from 1 through $n$:

$$\sum_{j=1}^{n} a_j = a_1 + a_2 + \cdots + a_n.$$

**Example 1.1.6** (The sum-of-squares formula in summation notation)**.** An important formula we will prove later in the course is

$$\sum_{j=1}^{n} j^2 = 1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

## 1.2 Introduction to counting

### 1.2.1 What is a counting problem?

A counting problem asks for the cardinality of a finite set. The first task is therefore to identify that set. We need to know:

- what one outcome is;
- when two outcomes are considered different;
- what restrictions an outcome must satisfy.

**Example 1.2.1** (Distinguished dice summing to ten)**.** Consider two ordinary six-sided dice, one red and one blue. How many ways can one obtain a roll whose sum is 10

Because the dice are distinguished by color, an outcome is an ordered pair $(r, b) \in \{1, 2, 3, 4, 5, 6\}^2$, where $r$ is the red result and $b$ is the blue result. We can list all of them

$$\{1,2,3,4,5,6\}^2 = \left\{ \begin{array}{llllll} (1,1), & (1,2), & (1,3), & (1,4), & (1,5), & (1,6), \\ (2,1), & (2,2), & (2,3), & (2,4), & (2,5), & (2,6), \\ (3,1), & (3,2), & (3,3), & (3,4), & (3,5), & (3,6), \\ (4,1), & (4,2), & (4,3), & (4,4), & (4,5), & (4,6), \\ (5,1), & (5,2), & (5,3), & (5,4), & (5,5), & (5,6), \\ (6,1), & (6,2), & (6,3), & (6,4), & (6,5), & (6,6) \end{array} \right\}$$

The outcomes whose sum is 10 are $(4, 6),\ (5, 5),\ (6, 4)$, so there are three.
Now let us change the problem in two different ways:

**Example 1.2.2** (What counts as a reroll outcome?)**.**

- **Variant 1** Consider two ordinary six-sided dice, one red and one blue. If a roll results in both dice having the same number, we reject the roll and try again.

How many ways can one obtain a roll whose sum is 10.

- **Variant 2** Consider two ordinary six-sided dice, one red and one blue. If a roll results in both dice having the same number, we reroll the red die until the two numbers are not the same. How many ways can one obtain a roll whose sum is 10.

It is not fully clear what the problem is asking us. If a ”way” records the entire history of rolls, then, in variant 1, rolling $(5, 5)$ and then rerolling and obtaining $(4, 6)$ is different from rolling $(4, 6)$. Actually,

arbitrarily many ties can occur before an accepted result, so there are infinitely many possible histories. Moreover, in variant 2, if one first rolls $(5, 5)$ there is actually no way to ever get 10 as a sum.
However, if we count only the final accepted ordered pair, then the problem becomes simple: $(5, 5)$ is not allowed, and the two outcomes with sum 10 are $(4, 6)$ and $(6, 4)$ just as before. This is true whether we reroll both dice (variant 1) after a tie or keep the red die and reroll only the blue die (variant 2).
This example shows that it is important to clearly understand what the problem, written in English, means at a precise mathematical level.
Next, we will develop tools that will help us solve the following challenge problem. You can try to solve it on your own, or come back to it once we covered the necessary counting techniques. For now, you should read and understand it, and use it as a motivating example for the things we will learn next

**Problem 1.2.3** (Committees with and without a chair)**.**

1. A company has 20 employees and they need to form a committee of 4 people. How many ways can this be done?
2. A company has 20 employees and they need to form a committee of 4 people: 1 chair and 3 other general committee members. How many ways can this be done?

### 1.2.2 The sum rule

Suppose one object is chosen from either of two finite sets $A$ and $B$. If the sets are disjoint, then every object in $A \cup B$ belongs to exactly one of the two sets. We may therefore add their cardinalities.

**Theorem 1.2.4** (Sum rule)**.** If $A$ and $B$ are finite disjoint sets, then

$$|A \cup B| = |A| + |B|.$$

**Example 1.2.5** (Lowercase letters or decimal digits)**.** A single character may be either a lowercase English letter or a decimal digit. How many allowed characters are there?
**Solution:** Let $L := \{\text{lowercase English letters}\}$, and $D := \{0, 1, \dots, 9\}$. There are $|L| = 26$ lowercase letters and $|D| = 10$ digits. No character is both a lowercase letter and a digit, so $L \cap D = \emptyset$. The sum rule gives

$$|L \cup D| = 26 + 10 = 36.$$

If uppercase English letters are also allowed, let $U$ be their set. The sets $L$, $U$, and $D$ are *pairwise disjoint*, and the number of allowed characters is

$$|L| + |U| + |D| = 26 + 26 + 10 = 62.$$

The same reasoning extends to any finite number of cases.

**Theorem 1.2.6** (General sum rule for disjoint sets)**.** Let $A_1, A_2, \dots, A_n$ be finite *pairwise disjoint* sets. Pairwise disjoint means that

$$A_i \cap A_j = \emptyset \qquad \text{whenever } i \neq j.$$

Then

$$\left| \bigcup_{j=1}^{n} A_j \right| = \sum_{j=1}^{n} |A_j|.$$

We can rephrase the sum rule in terms of outcomes of procedures, rather tnan sets.

**Theorem 1.2.7** (General sum rule for mutually exclusive cases). Suppose the outcomes of a task are divided into mutually exclusive cases $A_1, \ldots, A_n$, and every outcome belongs to exactly one case. The total number of outcomes is the sum of the numbers in the cases.

**Problem** (Check yourself). In the character example, identify $A_1$, $A_2$, and $A_3$ in the general sum rule, and verify pairwise disjointness.

**Disjointness is important!**
In the next example, let us see why checking disjointness is very important.

**Example 1.2.8** (Overlapping brands at two stores). Suppose Amazon sells three brands of juice and Walmart sells five brands. How many different brands can a shopper choose from if they may shop at either store?
**Wrong solution:** The number of brands at amazon is $|A| = 3$ and the number of brands at Walmart is $|W| = 5$, we need to find the total number of brands $|A \cup W|$. The sum rule tells us that

$$|A \cup W| = |A| + |W| = 3 + 5 = 8$$

The problem with this solution is that if the stores carry some of the same brands, adding gives duplicates. 8 is the correct answer only if $A$ and $W$ do not carry brands in common, that is if $A \cap W = \emptyset$. Depending on the overlap, the available information permits any total from 5 through 8.

**Warning 1.2.9** (Check the sum-rule hypotheses). Before using the sum rule, check that the cases are disjoint and that together they include every outcome being counted.

### 1.2.3 The product rule

Suppose an outcome is constructed through a fixed sequence of choices. If

- every complete sequence of choices produces exactly one outcome
- different choices produce different outcomes,
- the number of options at each step of the sequence of choices is fixed and does not depend on the choices made in the other steps,

then the total number of outcomes is obtained by multiplying the numbers of options at each step.

**Example 1.2.10** (Two-character lowercase strings). A two-character **string** is an ordered sequence $(c_1, c_2)$ in which each entry is one of the 26 lowercase letters `a-z`.
**Solution**: There are 26 choices for $c_1$ and, after any first choice, 26 choices for $c_2$. Therefore there are

$$26 \cdot 26 = 676$$

such strings. **Note**: The strings $af$ and $fa$ are different because their entries occur in different positions.

The argument allows multiple steps and also allows the specific choices at each step to depend on the other choices made. It is important that the total *number* of choices at each step remains the same.

**Theorem 1.2.11** (Product rule). Suppose every outcome is constructed uniquely by a fixed sequence of $m$ steps. Assume that, after any valid choices in the earlier steps, step $j$ has exactly $N_j$ valid options. Then the number of outcomes is

$$\prod_{j=1}^{m} N_j = N_1 N_2 \cdots N_m.$$

The available options may depend on the earlier choices; what must remain fixed is their **number** $N_j$.

**Example 1.2.12** (Three-letter strings with repetition). How many strings of length 3 made with letters `a-z` are there?

**Solution**: A string can be specified by first choosing the letter in position 1 (step 1), then the letter in position 2 (step 2), then in position 3 (step 3). For each of the steps there are 26 choices. Every string is obtained from exactly one sequence of such choices (different choices give different strings), so the product rule gives

$$26 \cdot 26 \cdot 26 = 26^3 = 17576.$$

**Example 1.2.13** (Three-letter strings without repetition). How many strings of length 3 made with letters `a-z` are there such that no letter appears more than once?

**Solution**: A string can be specified by first choosing the letter in position 1 (step 1), then the letter in position 2 (step 2) and making sure that it is not the same as the letter chosen in position 1, then choosing a letter in position 3 (step 3) while making sure that it is not the letter chosen in position 1 nor in position 2.

- Step 1: 26 choices
- Step 2: 25 choices (26 letters except the one chosen in step 1)
- Step 3: 24 choices (26 letters except the one chosen in step 1 and the one chosen in step 2)

Every valid string is obtained from exactly one sequence of such choices (different choices give different strings), so the product rule gives

$$26 \cdot 25 \cdot 24 = 15600.$$

It is important to note that in this example the *specific* choices at steps 2 and 3 depend on the choices made in the previous step but the *number* of possible choices does not! This allows us to apply the product rule.

**Example 1.2.14** (Three-letter strings with no symbol appearing three times). How many strings of length 3 made with letters `a-z` are there such that no letter appears more than twice?

**Wrong/Incomplete solution**: A string can be specified by first choosing the letter in position 1 (step 1), then the letter in position 2 (step 2) (without any restrictions, because the second letter can be the same as the first one and not lead to a forbidden outcome)
In step 3 we choose the letter for position 3.

- Step 1: 26 choices
- Step 2: 26 choices
- Step 3: ?? choices (it depends, if in step 1 and 2 we chose the same letter then we cannot choose it again so we have 25 choices; if we did not choose the same letter we can choose any letter, so 26 choices. )

**Possible correct solution**: A string can be specified by choosing by first choosing the letter in position

1 (step 1), then the letter in position 2 (step 2), then the letter in position 3 (step 3), all while making sure not to choose a string with the same letter repeated 3 times.
Let us distinguish 2 cases:

- *D:* In step 1 and step 2 we choose different letters
- *S:* In step 1 and step 2 we choose the same letter

By the sum rule, the total amount of strings is going to be $|D \cup S| = |D| + |S|$ since $D \cap S = \emptyset$ (we cannot both select the same and different letters in step 1 and 2).
Let us compute $|D|$. Since the first two letters are different we have the following options:

- Step 1: 26 choices
- Step 2: 25 choices (cannot be the same as step 1)
- Step 3: 26 choices (since the first are different, there are no restrictions)

so $|D| = 26 * 25 * 26$ by the product rule
Let us compute $|S|$. Since the first two letters are the same we have the following options:

- Step 1: 26 choices
- Step 2: 1 choice (must be the same letter as step 1)
- Step 3: 25 choices (since the first two letters are the same we cannot choose that letter again)

so $|S| = 26 * 1 * 25$
The final answer is $26 * 25 * 26 + 26 * 1 * 25 = 17550$.

## 1.3 Strings: an important type of counting problems

The preceding examples illustrate an important type of counting problem: counting ordered sequences of symbols.

**Definition 1.3.1** (String over an alphabet). Let $\Sigma$ be a finite set of symbols that we call an alphabet. A **string of length** $k$ over an alphabet $\Sigma$ is an ordered sequence

$$(a_1, a_2, \dots, a_k),$$

where each $a_j \in \Sigma$.
Using set notation, a string of length $k$ is an element of the Cartesian product $A^k$.

The position of each symbol is important. For example, `ab` and `ba` are different strings. Two strings are equal precisely when they have the same symbol in every position. Repetition is allowed unless the problem states otherwise.
There is also one string of length 0, called the **empty string**. It contains no symbols.

**Theorem 1.3.2** (Strings with repetition). Let $\Sigma$ be an alphabet with $n \geq 1$ symbols, and let $k$ be a nonnegative integer. The number of strings of length $k$ over $A$ is $n^k$.

*Proof.* There are $k$ positions. Each position may contain any of the $n$ symbols, independently of the symbols chosen in the other positions. The product rule therefore gives

$$\underbrace{n \cdot n \cdots n}_{k \text{ factors}} = n^k.$$

When $k = 0$, the product has no factors and is equal to 1, corresponding to the one empty string. $\square$

**Example 1.3.3** (Three-digit strings with repetition)**.** Let $D = \{0, 1, \dots, 9\}$ be the alphabet of decimal digits. The number of strings of length 3 over $D$ is

$$10^3 = 1000.$$

These are strings rather than three-digit positive integers, so a string such as `042` is included.

Suppose now that no symbol may occur more than once.

**Theorem 1.3.4** (Strings without repetition)**.** Let $\Sigma$ be an alphabet with $n$ symbols, and let $k$ be a nonnegative integer. The number of strings of length $k$ over $\Sigma$ with no repeated symbol is

$$P(n, k) := n(n-1)\cdots(n-k+1).$$

This formula makes sense when $0 \le k \le n$, the product in the first line has $k$ factors.
When $k = 0$ we have $P(n, 0) = 1$. When $k > n$ then $P(n, k) = 0$.

*Proof.* Assuming that $0 \le k \le n$. There are $n$ choices for the first position. Once a symbol has been used, it cannot be used again, so there are $n - 1$ choices for the second position, $n - 2$ choices for the third position, and, finally, $n - k + 1$ choices for the $k$th position. The product rule gives

$$P(n, k) = n(n-1)\cdots(n-k+1).$$

The specific digits available at each step depend on the earlier choices, but the number of available digits does not. This is exactly what the product rule requires.
If $k > n$, a string without repetition would require more than $n$ distinct symbols from an alphabet containing only $n$ symbols. No such string exists, so $P(n, k) = 0$. $\square$

**Example 1.3.5** (Three-digit strings without repetition)**.** Let $D = \{0, 1, \dots, 9\}$ be the alphabet of decimal digits. The number of strings of length 3 with no repeating digits over $D$ is

$$P(10, 3) = 10 \cdot 9 \cdot 8 = 720.$$

These are strings rather than three-digit positive integers, so a string such as `042` is included.

## 1.4 External References

### 1.4.1 Additional material (optional)

- Counting Rocks! - Basics of Set Theory
- Discrete Math II - 6.1.1 The Rules of Sum and Product
- Counting Rocks! - The addition, subtraction, and multiplication principles